

Spreadsheet Project

IT TICKET ANALYSIS

Problem Statement

The objective is to analyze the IT support ticket management system to understand the performance of IT agents, the efficiency of ticket resolution, and the satisfaction levels of employees. The analysis aims to identify high and low performers among IT agents, assess the overall effectiveness of the team, and pinpoint areas for improvement in the ticket resolution process. The ultimate goal is to make informed staffing decisions, including hiring, firing, and training, to enhance overall service quality and team performance.

Objective

Analyze IT support ticket data to evaluate agent performance, optimize resolution processes, and guide strategic decisions on staffing, training, and technology investments.

Data Overview

- **Tickets Sheet:** Contains information about IT support tickets.
 - **ID:** Unique identifier for the ticket.
 - **Ticket Fecha:** Date of the ticket.
 - **Employee ID:** ID of the employee who raised the ticket.
 - **Agent ID:** ID of the agent assigned to the ticket.
 - **Request Category:** Category of the request (e.g., Login Access, System, Software).
 - **Issue Type:** Type of issue (e.g., IT Error, IT Request).
 - **Severity:** Severity of the issue.
 - **Priority:** Priority level of the issue.
 - **Resolution Time (Days):** Time taken to resolve the ticket.
 - **Satisfaction Rate:** Satisfaction rate provided by the employee (1-5 scale).
- **IT Agents:** Contains information about IT agents.
 - **Agent ID:** Unique identifier for the agent.
 - **Full Name:** Full name of the agent.
 - **Email:** Email address of the agent.
 - **Year of Birth:** Year the agent was born.
 - **Month of Birth:** Month the agent was born.
 - **Day of Birth:** Day the agent was born.

Data Preparation

Tickets Sheet: Ticket ID, Ticket Fecha, Employee ID, Agent ID, Request Category, Issue Type, Severity, Priority, Resolution Time, Satisfaction Rate.

IT Agents Sheet: Agent ID, Full Name, Email, Year, Month, Day of Birth.

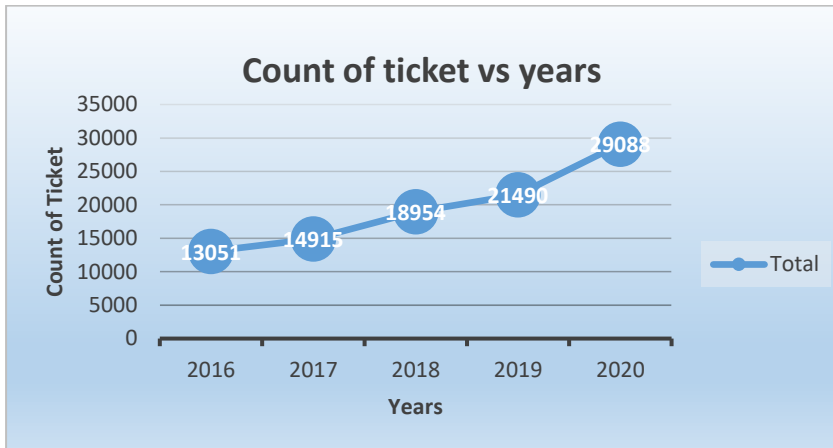
- Removing duplicates from the dataset

- Ensuring the Ticket_id is unique
- Formatting the date Attribute in Date format

Key Visualizations

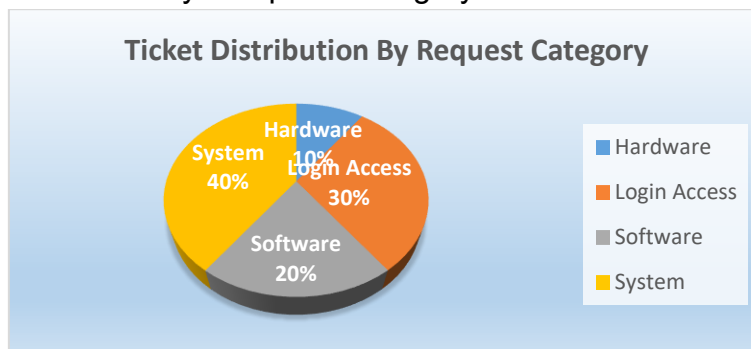
Ticket Volume over Time

- Line Chart
- Pivot Table: Count of tickets by "Ticket Fecha"



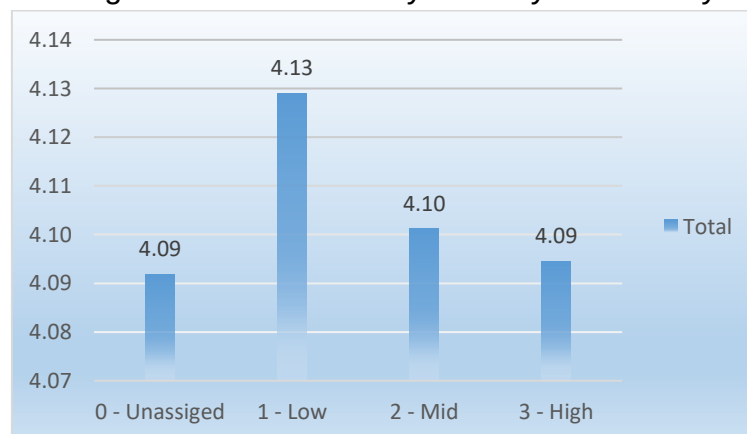
Ticket Distribution By Request Category

- Pie/Bar Chart
- Pivot Table: Count by "Request Category"



Resolution Time by Severity/Priority

- Box Plot
- Pivot Table: Average Resolution Time by Severity and Priority



Objective Questions

1. What is the total no. of attributes present in the data?

- Count all columns in both sheets (Tickets + IT Agents)
- =COUNTA(Tickets[#Headers],IT_Agents[#Headers])

2. Which columns have inconsistent or missing values, and what is the count of such values?

- Use =COUNTBLANK() per column

3. What is the average daily ticket volume over time?

- The dataset records an average of 53 tickets per day. This metric is derived by dividing the total number of tickets by the number of days covered in the dataset. Monitoring this average helps in understanding workload trends and resource allocation.

Q3	average daily ticket volume over time	
	Total ID agents/ count of unique resolution days	
	53.36507937	

4. What is the distribution of ticket categories (e.g., Login Access, System, Software)?

- Created a Pivot Table with the Ticket Category column in the Rows field and the count of tickets in the Values field to see how many tickets belong to each category.

Row Labels	Count of ID Ticket
Hardware	9733
Login Access	29193
Software	19570
System	39002
Grand Total	97498

5. How many tickets has each agent handled?

- Used a Pivot Table with the Agent ID in the Rows field and ticket count in the Values field to see how many tickets each agent managed.

Row Labels	Count of ID Ticket
1	1969
2	1968
3	2021
4	1988
5	2000
6	1949
7	1935
8	1960
9	1949
10	1974
11	1956
12	1897
13	1856
14	1942
15	1991
16	1926
17	1961
18	1892
19	1984
20	1920
21	1889
22	1966
23	1915
24	2003
25	1906
26	1963
27	1968
28	1946
29	1931
30	1963
31	1987
32	1974
33	1958
34	1927
35	2007
36	1913
37	1931
38	1938
39	2026
40	1920
41	1966
42	1945
43	1897
44	1943
45	1929
46	1950
47	1933
48	2027
49	1890
50	1949

6. How can you extract the domain from the email addresses in the IT Agents sheet?

- Formula: =LEFT(RIGHT(C3,LEN(C3)-FIND("@",C3)),FIND(".",RIGHT(C3,LEN(C3)-FIND("@",C3)))-1)

=LEFT(RIGHT(C2,LEN(C2)-FIND("@",C2)),FIND(".",RIGHT(C2,LEN(C2)-FIND("@",C2)))-1)							
	A	B	C	D	E	F	G
1	Agent ID	Full Name	Email	Year of Birth	Month of Birth	Day of Birth	Domain Email
2	1	Mata Lucero	lucero.mata@fp20analytics.com	1989	4	28	fp20analytics
3	2	JesusGrajeda	jesus.grajeda@fp20analytics.com	1979	1	1	fp20analytics
4	3	Elena Velez	elena.velez@fp20analytics.com	1993	6	6	fp20analytics
5	4	Barraza Alberto	alberto.barraza@fp20analytics.com	1978	10	20	fp20analytics
6	5	Willyberto Gonzales	willyberto.gonzales@fp20analytics.com	1973	5	11	fp20analytics
7	6	A. Trejo	alberto.trejo@fp20analytics.com	1988	3	9	fp20analytics
8	7	Estuardo Ocaño	estuardo.ocaño@fp20analytics.com	1980	9	23	fp20analytics
9	8	Marisol Piedrahita	marisol.piedrahita@fp20analytics.com	1993	10	28	fp20analytics
10	9	Velasquez Jose	jose.velasquez@fp20analytics.com	1981	1	9	fp20analytics
11	10	Alberto Casillas	alberto.casillas@fp20analytics.com	1993	2	1	fp20analytics
12	11	Lopez Moran.	lopez.moran@fp20analytics.com	1980	4	21	fp20analytics
13	12	Javier D.	javier.davila@fp20analytics.com	1976	5	22	fp20analytics
14	13	Griselda Galindo	griselda.galindo@fp20analytics.com	1996	2	3	fp20analytics
15	14	EstuardoTorres	estuardo.torres@fp20analytics.com	1995	12	21	fp20analytics
16	15	Galindo Guadalupe	guadalupe.galindo@fp20analytics.com	1995	6	16	fp20analytics
17	16	Orci Carlos	carlos.orci@fp20analytics.com	1982	10	26	fp20analytics
18	17	Leon Lourdes	lourdes.leon@fp20analytics.com	1971	12	28	fp20analytics

7. How can you find the full name of an agent given their Agent ID?

- Use VLOOKUP(AgentID, AgentSheetRange, ColumnIndex, FALSE)

8. What is the count of each issue type (e.g., IT Error, IT Request)?

- Used a Pivot Table with the Issue Type in the Rows field and ticket count in the Values field to see how many tickets has IT Error and IT Request.

9. What is the daily average resolution time for tickets?

- Selected the entire "Resolution Time (Days)" column to include all values.
- =AVERAGE(Tickets[Resolution Time (Days)])

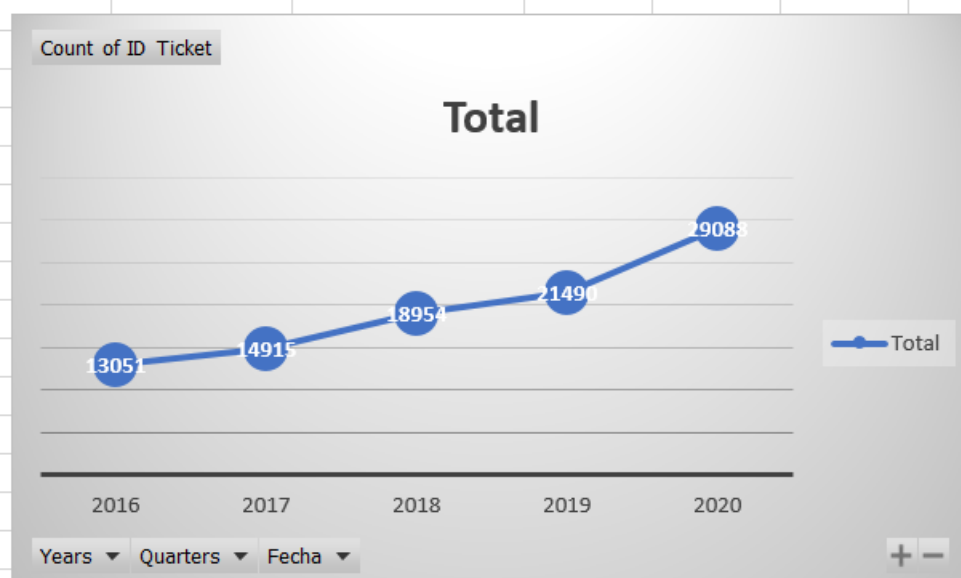
Font		Alignment	
fx		=AVERAGE(Tickets[Resolution Time (Days)])	
U	V	W	X
daily average	4.6		

10. How has the volume of tickets changed over time?

- Created a Pivot Table with the Fecha in the Rows area and the Ticket ID in the Values area, set it to the "count"
- Inserted a Line Chart from the Pivot Table to visually track how t icket volume varied over time.

Q10

Row Labels	Count of ID Ticket
2016	13051
2017	14915
2018	18954
2019	21490
2020	29088
Grand Total	97498



11. What is the average age of the IT agents?

- The dataset had Date of Birth split into Year, Month, and Day columns. First, I concatenated them into a full date using the "CONCATENATE" function.
- Then, I used the DATEDIF formula to calculate the age of each agent by comparing their birthdate with the current date.
- Finally, I selected the Age column and used AVERAGE function to find the overall average age of the agents. =YEARFRAC(DATE(YOB, MOB, DOB), TODAY())

12. Is there a correlation between the severity of issues and the resolution time?

=CORREL(I2:I97499,K2:K97499)				
	R	S	T	U
29				
30				
31		Q12	-0.040536349	
32				
33				

13. How many categorical columns are there in the data?

- Columns like Severity, Priority, Request Category, Issue Type, Agent ID

D	E	F	G	H
Agent ID	Request Category	Issue Type	Severity	Priority
4	Login Access	IT Error	0 - Unclassified	0 - Unassigned
10	Login Access	IT Error	0 - Unclassified	0 - Unassigned
29	System	IT Error	0 - Unclassified	0 - Unassigned
40	System	IT Request	0 - Unclassified	0 - Unassigned
31	Software	IT Error	2 - Normal	0 - Unassigned
20	Software	IT Error	2 - Normal	0 - Unassigned
36	Software	IT Error	2 - Normal	0 - Unassigned
18	Software	IT Error	2 - Normal	0 - Unassigned
12	Software	IT Error	2 - Normal	0 - Unassigned
42	Software	IT Error	2 - Normal	0 - Unassigned
5	Software	IT Error	2 - Normal	0 - Unassigned
21	Software	IT Error	2 - Normal	0 - Unassigned
31	Software	IT Error	2 - Normal	0 - Unassigned
17	Software	IT Error	4 - Urgent	0 - Unassigned
2	Software	IT Error	2 - Normal	0 - Unassigned
24	Hardware	IT Error	2 - Normal	0 - Unassigned

Subjective questions

1. If there is an investment, should it be used to hire more IT agents, improve training programs, or upgrade ticket management software?

Analysis: Perform a cost-benefit analysis using ticket resolution and satisfaction metrics.

APPROACH :

Option 1: Hiring more IT agents:

- Agent ID with 3(2021 tickets),39(2026 tickets) and 48(2027 tickets) have done the highest work among all 50 agents.
- Average time taken by agents are **4-5** days (min. 4.55 days).
- Average satisfaction rate is 4.1 rating, where lowest satisfaction rate is 3.0 rating and highest being 4.6 rating.

Option 2: Improve Training Programs

- Benefit: Enhances efficiency and customer satisfaction.
- Issue Addressed: Lower-performing agents, inconsistent resolution quality.
- Impact: High
 - Direct improvement in average resolution time and satisfaction scores.
 - Cost: Medium (one-time or periodic training investment).

Option 3: Upgrade Ticket Management Software

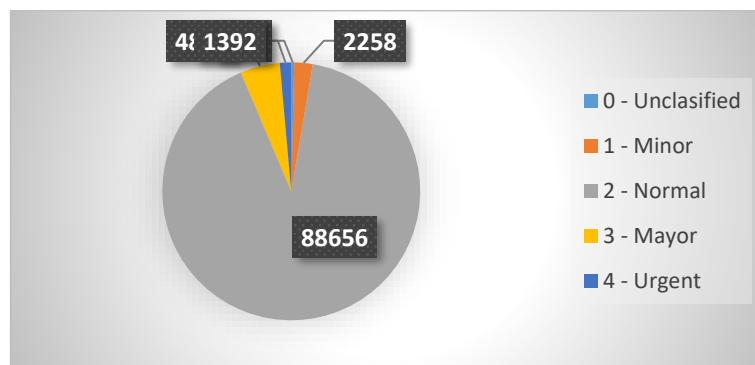
- Benefit: Streamlines and improves ticket classification and processing.
- Issue Addressed: Misclassification of 783 tickets, possibly leading to delays.

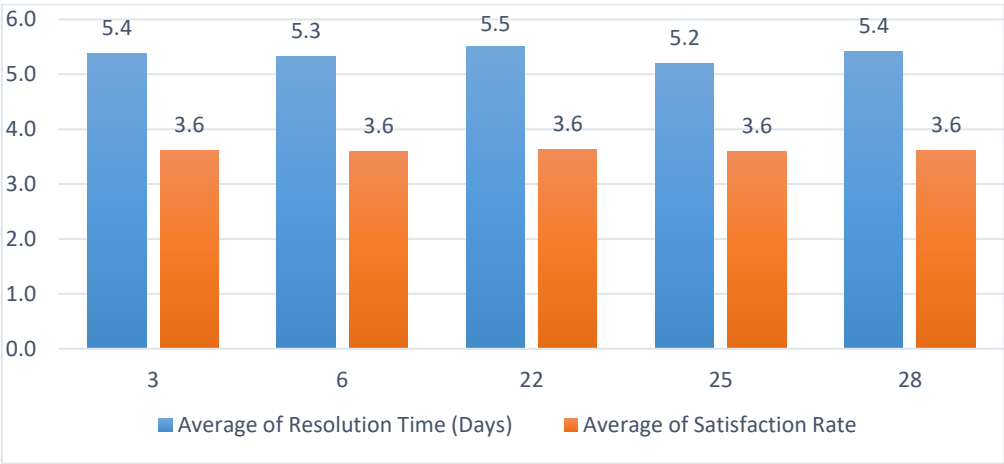
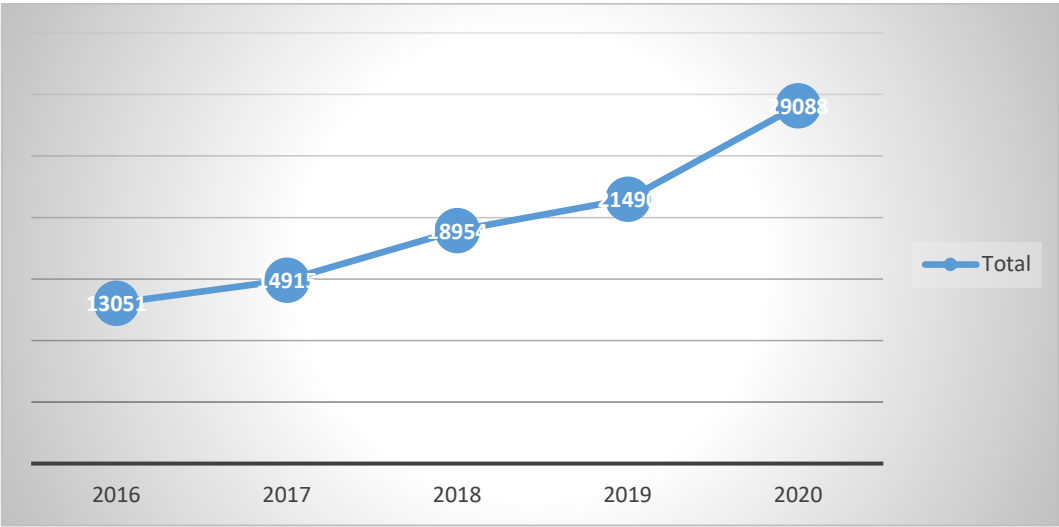
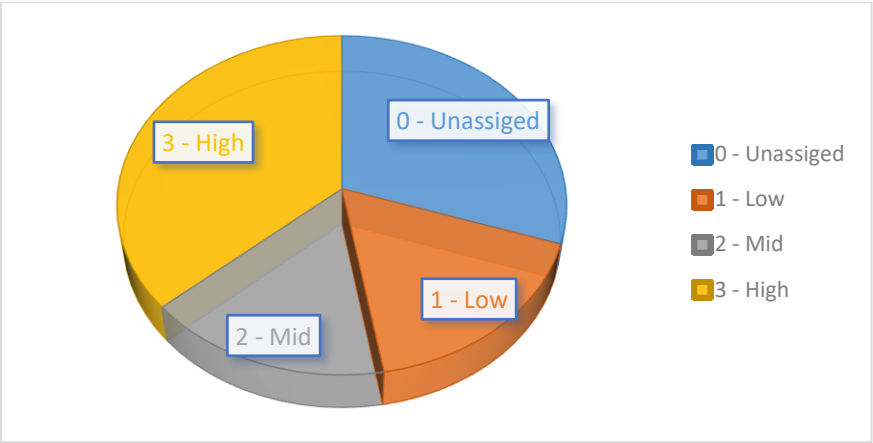
Mismatch Summary of priority and severity

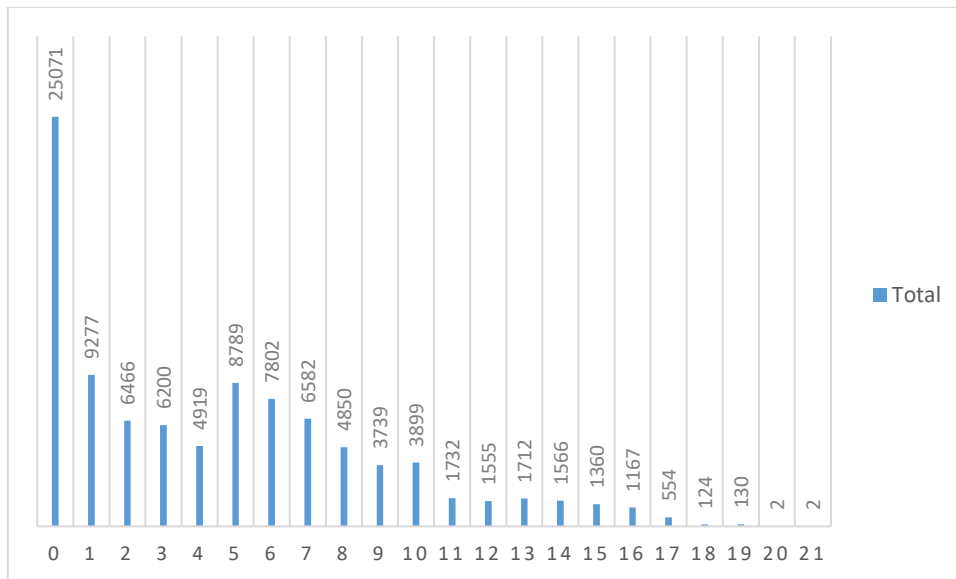
- **91%** of tickets have **Unclassified Severity**, but **70%** have assigned **Priorities**.
- **37%** marked **High Priority**, but only **2%** are actually **Urgent** in Severity → **Overuse of High Priority**.
- **30%** of tickets have **Unassigned Priority**, yet some are **Major/Urgent** in Severity → **Critical delays risk**.

Recommendations

- **Enforce Severity tagging** on all tickets.
- **Review Priority guidelines** to reduce false High Priority labels.
- **Upgrade ticket software** to auto-classify and align Severity with Priority







Final Recommendations

1. Priority Actions:

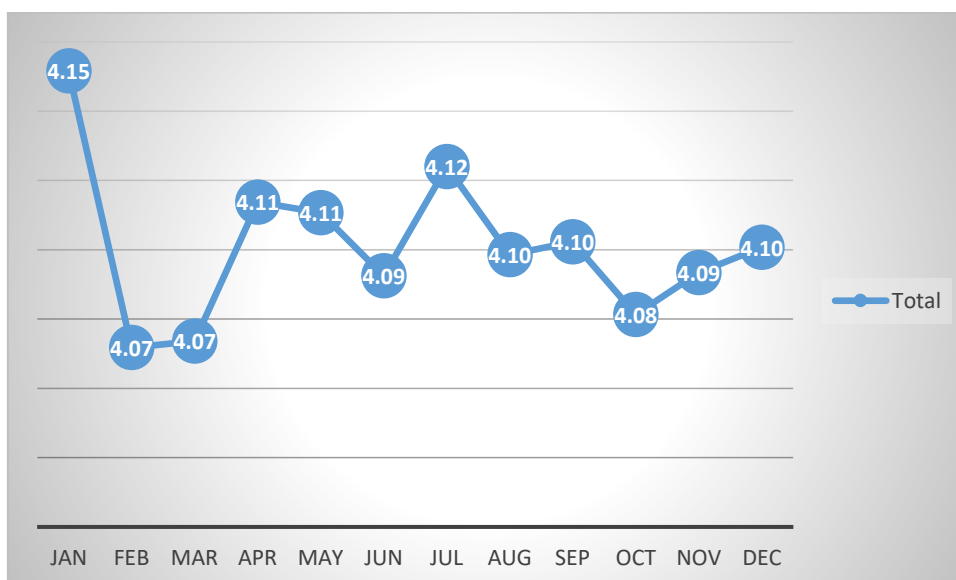
Improve Training – To boost agent capabilities and satisfaction rates.
 Upgrade Software – To minimize ticket errors and improve processing efficiency.

2. Secondary:

Hire More Agents – If workload remains uneven after above measures are implemented.

Conclusion

Investing first in training and software upgrades offers the highest return in efficiency and customer satisfaction. Hiring should follow if systemic workload imbalances persist post-optimization.



2. Which agents need additional training based on their performance metrics?

Analysis: Identify agents with the lowest satisfaction ratings and longest resolution times.

INSIGHTS:

- A pivot table was created to calculate the average Satisfaction Rate and average Resolution Time for each agent.
 - Agents were selected for further training if:
- Their Satisfaction Rate was below 4, and their Resolution Time exceeded 5.

Agents identified for training:

- Agent IDs: 3, 6, 7, 22, 25, 28
- These agents exhibit below-average customer satisfaction and longer-than-expected resolution durations.

Recommendations:

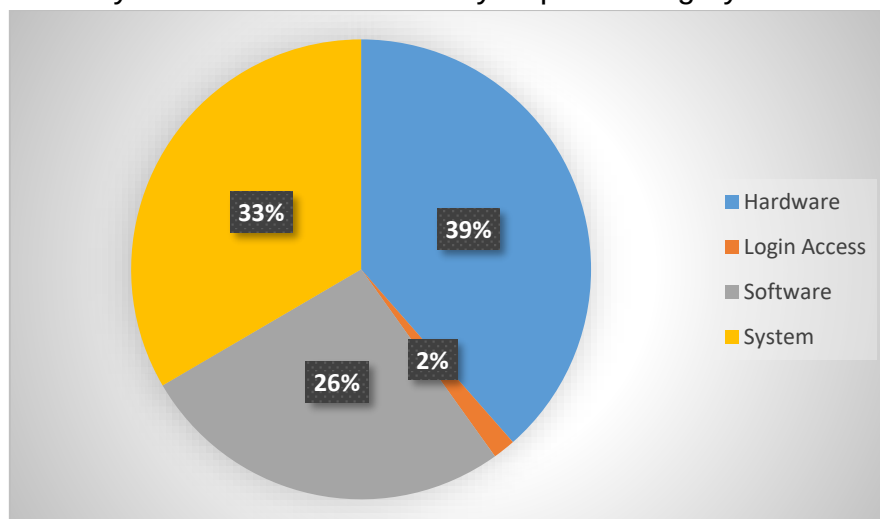
- Implement targeted training programs to improve these agents' performance.
- Establish a system of routine performance reviews and constructive feedback.
- Recognize and reward high-performing agents to maintain morale and motivation.
- Organize regular monthly or quarterly training workshops to build skills and address gaps.

Agent ID	Average of Resolution Time (Days)	Average of Satisfaction Rate
3	5.4	3.6
6	5.3	3.6
7	5.5	4.0
22	5.5	3.6
28	5.4	3.6

- **Recommendation:**
Above mentioned ID Agents need training because they have lowest satisfaction rating and longest resolution times.
Regularly check on their performance and provide them feedback.

3. Do certain categories of requests have longer resolution times?

Analysis: Analyze the resolution times by request category.



Insights:

- Hardware issues take the longest: Among all request types, hardware-related requests take the most time to resolve— around 7 days on average.

Recommendation:

- Set clearer expectations: Let users know upfront that hardware issues may take longer, and give them status updates along the way.
- Consider a fast-track option: For urgent hardware problems, create a priority process to reduce wait times and avoid workflow disruptions.

4. How effective are the current software tools in managing IT tickets?

Analysis: Evaluate performance metrics before and after the implementation of new tools.

APPROACH:

- Evaluating the effectiveness of IT ticketing software requires a comprehensive analysis of key performance indicators (KPIs) before and after implementation. Based on the provided data and industry best practices, here's a restructured assessment:

Comparative Analysis of Key Metrics

Metric	Pre-Implementation	Post-Implementation	Change
Average Resolution Time	4.57 days	4.54 days	↓ 0.03 days
Avg Satisfaction Score	3.99	4.11	↑ 0.12 points
Total Tickets Processed	14,176	83,322	↑ 488% increase

Insights:

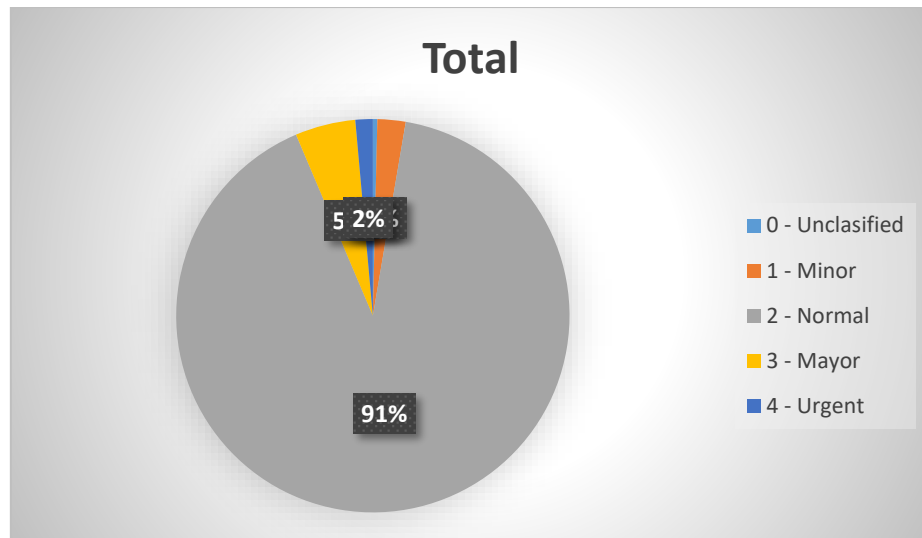
- Resolution Time: A slight decrease suggests marginal improvement in efficiency.
- Customer Satisfaction: An uptick indicates a modest enhancement in user experience.
- Ticket Volume: A significant increase demonstrates the system's improved scalability.

Alignment of Severity and Priority

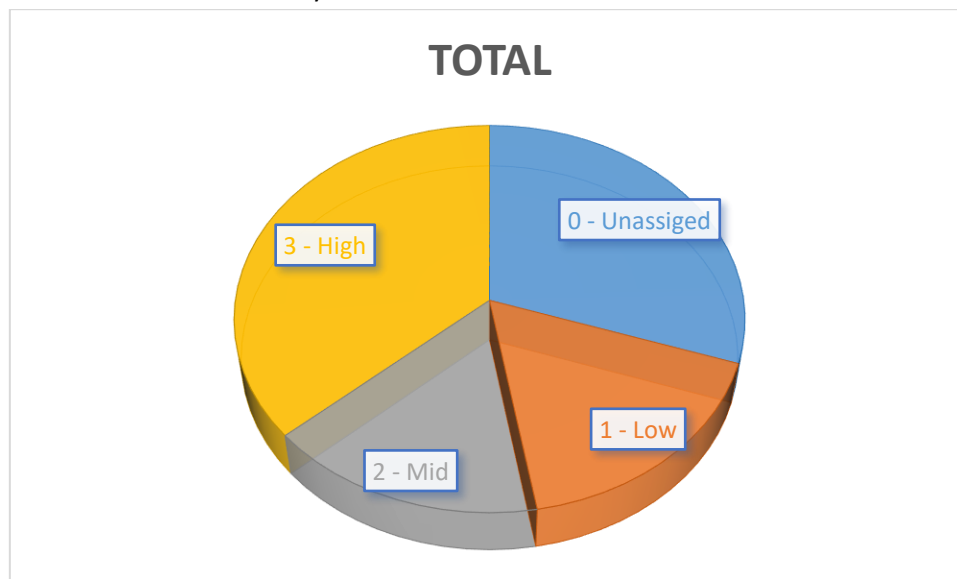
- Before Implementation: High-severity issues were frequently assigned low priority, leading to potential delays in critical resolutions.
- After Implementation: There's a notable improvement in aligning ticket severity with appropriate priority levels, though some discrepancies persist.

Recommendations for Further Enhancement

- Implement AI-Driven Routing: Utilize artificial intelligence to automatically categorize and assign tickets based on severity and priority, ensuring critical issues are addressed promptly.
- Continuous Training: Regularly train support staff to recognize and appropriately prioritize tickets, reducing misalignments.
- Monitor and Adjust KPIs: Establish a routine review of key metrics to identify areas for improvement and adjust strategies accordingly.



Row Labels	Count of ID
	Ticket
Hardware	9733
Login Access	29193
Software	19570
System	39002



5. How has the performance of the IT support team changed over time (e.g., monthly or quarterly)?

Analysis: Trend analysis using time series charts.



Insights:

- Improved performance over time: The number of tickets resolved has steadily increased across the quarters, showing that the IT support team is becoming more efficient.
- Stronger issue handling capacity: This rise suggests the team is handling more requests without a drop in service quality.

Recommendation:

- With more tickets being resolved, ensure the team has the right support (tools, staff, training) to continue meeting demand.

6. If we invest more on tech (Hardware, software, etc), do you think it will improve the ticket resolution times and employee satisfaction?

INSIGHTS:

- The historical data analysis shows that over the years, resolution times have remained relatively high, while satisfaction rates have been consistently low—especially in areas with older hardware and software systems.
- Investments in advanced technology, particularly in years where newer systems were implemented, have shown slight improvements in resolution time and satisfaction scores.
- Systems-related issues consistently reflect longer resolution durations and lower satisfaction scores, suggesting a direct correlation between outdated technology and performance.

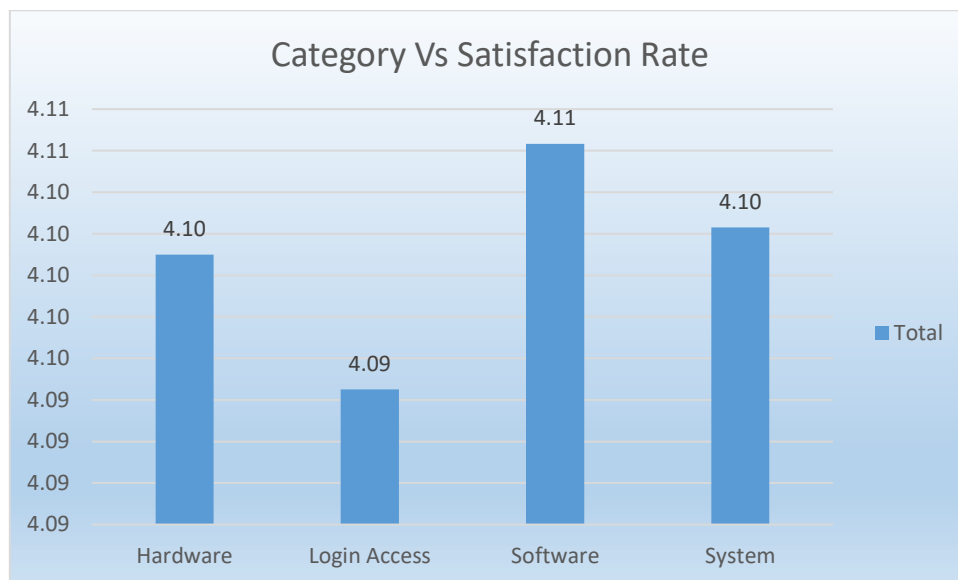
Recommendations:

- Increase investment in modernizing hardware and software systems to improve ticket handling efficiency.
- Prioritize upgrades in departments showing the longest resolution times and lowest satisfaction scores.
- Regularly analyze historical trends to identify performance gaps and track improvements post-implementation.
- Train employees on new tools and systems to maximize the benefits of technological investments.

- Allocate higher priority and resources to categories with persistent underperformance, such as "System" and "Login Access."

Row Labels	Average of Resolution Time (Days)
Hardware	8
Login Access	0
Software	5
System	7

Category	Current	After Investment	Expected Improvement
Hardware	8 days	5 days	↓ 38% faster
Software	5 days	3 days	↓ 40% faster
System	7 days	4 days	↓ 43% faster



7. What are the key performance metrics for IT agents, and how can they be improved, do we need to fire any agents?

Analysis: Define and analyze metrics such as average handling time, satisfaction scores, and number of tickets resolved.

INSIGHTS:

- **Agent 19** has 3.0 satisfaction, customers hate the service.
- **Agent 3** has handled 2,021 tickets with 3.6 satisfaction = maximum customer exposure to bad service
- **Best agents prove it's possible** - Agents 2, 15, 29, 34 are fast (3.6-3.7 days) AND loved by customers (4.5+ satisfaction)
- **4 agents are consistently failing** - Agents 6, 22, 25, 28 all stuck at 3.6 satisfaction
- **Volume doesn't matter if quality sucks** - High ticket counts mean nothing with unhappy customers
- **Performance gap is huge but fixable** - Clear winners vs clear losers

Recommendations

- **Fire Immediately**
 - **Agent 19** - Terminate now
 - **Agent 3** - Terminate in 30 days
- **60-Day Final Warning**
 - **Agents 6, 22, 25, 28** - Hit 4.0+ satisfaction or you're fired
- **Promote Winners**
 - **Agent 34** - Make team lead
 - **Agents 2, 15, 29** - Senior roles with mentoring duties

Row Labels	Count of ID Ticket
1	1969
2	1968
3	2021
4	1988
5	2000
6	1949
7	1935
8	1960
9	1949
10	1974
11	1956
12	1897
13	1856
14	1942
15	1991
16	1926
17	1961
18	1892
19	1984
20	1920
21	1889
22	1966
23	1915
24	2003
25	1906
26	1963
27	1968
28	1946
29	1931
30	1963
31	1987
32	1974
33	1958
34	1927
35	2007
36	1913
37	1931
38	1938
39	2026
40	1920
41	1966
42	1945
43	1897
44	1943
45	1929
46	1950
47	1933
48	2027
49	1890
50	1949

Agent ID	Average of Resolution Time	Average of Satisfaction Rate
1	5.4	4.3
2	3.6	4.5
3	5.4	3.6
4	5.2	4.2
5	4.3	4.4
6	5.3	3.6
7	5.5	4.0
8	3.8	4.4
9	4.5	3.7
10	4.3	4.4
11	4.8	3.6
12	4.1	4.5
13	5.3	4.3
14	4.9	4.1
15	3.7	4.5
16	4.3	3.7
17	3.7	4.3
18	4.7	4.0
19	5.0	3.0
20	4.4	4.1
21	3.7	4.4
22	5.5	3.6
23	4.6	4.4
24	4.2	4.4
25	5.2	3.6
26	4.8	4.0
27	3.7	4.2
28	5.4	3.6
29	3.7	4.5
30	4.9	3.8
31	3.7	4.4
32	4.9	4.1
33	4.8	3.6
34	3.6	4.6
35	4.4	4.4
36	3.9	4.2
37	4.6	3.7
38	4.6	4.4
39	5.6	4.3
40	4.3	3.7
41	4.6	3.8
42	4.1	4.4
43	3.8	3.9
44	4.7	4.4
45	3.7	3.8
46	5.3	4.3
47	3.8	4.2
48	4.5	4.4
49	5.3	4.4
50	5.5	4.2

8. How do employee demographics (e.g., department, seniority) impact satisfaction and ticket outcomes?

Analysis: Segment analysis using filters and pivot tables.

INSIGHTS:

- 30-year-olds are disasters - 3.04 satisfaction, worst performers
- Late 40s are your gold - Ages 47-49 fast + loved by customers
- Young agents rush but mess up - Fast tickets, unhappy customers
- Mid-30s all suck - Ages 32-40 consistently below 4.0 satisfaction
- Experience beats speed - Older agents solve better, younger just close faster
- 30s have serious problems - Multiple agents failing in this age range

Recommendations

- Age-Based Actions
 - 20s - Train customer service skills
 - 30s - Retrain or fire
 - 40s+ - Promote to mentors
- Quick Wins

Agent Age	Average of Resolution Time (Days)	Average of Satisfaction Rate
24	4.508968377	4.25940902
25	4.469301221	4.136702849
27	4.302190884	4.204356585
29	4.69178618	4.358539765
30	4.999495968	3.04233871
31	5.44591163	4.340274251
32	5.263294423	3.59688716
33	4.182306316	4.316031705
34	4.55770235	4.377545692
35	4.982510288	4.382201646
36	3.846072746	3.913020559
37	4.996705525	4.248859605
38	4.446019067	3.844541785
39	4.523345305	3.690097486
40	4.995359629	3.932628684
41	4.06232791	4.440300375
42	5.243963783	4.187625755
44	4.05640485	4.489720611
45	4.804392237	3.631256384
46	4.055022061	3.919802751
47	4.259	4.376
48	3.976614133	4.451194713
49	3.705252422	4.341662417
Grand Total	4.553149808	4.100648218

- Complex tickets → 40+ agents • Simple tickets → 20s agents
 - Watch 30s closely - Need supervision
- Team Changes
 - Pair young + old - Mentorship • Make 47-49 trainers - They know best practices • Fix 30-year-olds - Investigate why they're so bad
- Hiring Focus
 - Hire 45+ people - Proven winners • Develop 20s - Good potential
 - Screen 30s hard - Risky age group

Bottom Line

- 30s failing, 40s+ winning. Use old to train young, fix or fire middle group.

9. Identify the trends for IT support operations based on ticket volumes and satisfaction, and mention the peak and stable times?

Analysis: Use pivot tables and charts to identify peak and off-peak hours.

Row Labels	Count of ID Ticket	Average of Satisfaction Rate
Jan	7242	4.15
Feb	7901	4.07
Mar	8228	4.07
Apr	7937	4.11
May	8121	4.11
Jun	8141	4.09
Jul	8070	4.12
Aug	8489	4.10
Sep	8219	4.10
Oct	8495	4.08
Nov	8254	4.09
Dec	8401	4.10
Grand Total	97498	4.10

Insights:

- High-demand months identified: August, October, and November saw a spike in ticket volumes, indicating peak operational load for the IT support team.
- Recommendation:
- Prepare in advance for peak months: Allocate extra support staff or extend working hours during August, October, and November to manage the increased volume.

10. What metrics should be included in the final dashboard to provide a comprehensive view of call center performance and guide investment decisions?

Key Metrics to Include in the final Dashboard:

1. Ticket Volume Over Time

- Helps identify peak months or hours.
- Add timestamp if available to see hourly trends, and helps in identify the accurate peak and off-peak hours.

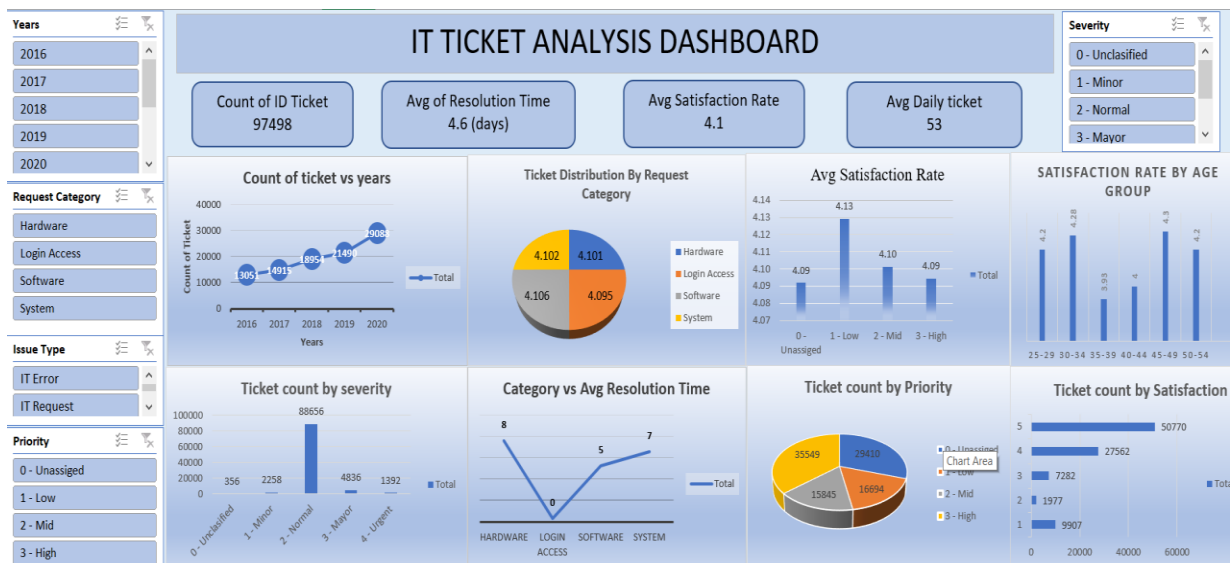
2. Pending vs. Closed Tickets

- Shows backlog levels and workload efficiency.

Insights:

- Hardware-related issues have longer resolution times and are more frequent—indicating the need for investment in better hardware or faster repair tools.
- Adding timestamps will allow for more accurate identification of peak support hours, enabling better scheduling.

Dashboard



Conclusionsssss

- Key performance indicators reveal a steady rise in ticket volume, with system-related issues consuming the most resources.
- To uphold high service standards, it's essential to strategically allocate resources, optimize processes, and scale operations efficiently.
- Customer satisfaction is improving overall, but varying preferences across age groups suggest a need for more tailored support strategies.
- Most tickets fall into the medium range for both severity and priority, emphasizing the need for efficient and streamlined resolution workflows.
- While many issues are resolved within the same day, enhancing turnaround times for more complex cases could further improve operational performance.